

Trading Strategy Workshop

Workshop Day 1 – Build Your First Trading Strategy

Today, you'll work in teams to build the first version of your trading system. You'll explore data, build a simple strategy, simulate trades, and visualize results. Focus on correctness and understanding—speed comes later.

Load & Explore Stock Data

- Load the provided CSV files containing stock price history (AMD, META, MSFT, SBUX).
- Visualize how each stock's price changed over time.

Checkpoint

Does the chart match what you expect for recent stock movements?

Implement Your First Strategy

- Write a simple momentum-based strategy
 - $\text{signal}(t+1) = 1$ if $P(t) > P(t-1)$, else 0
- Use this signal to predict price movement for each day.
- Compare the predicted direction vs actual returns.

Checkpoint

Compute and plot the **mean squared error (MSE)** and **correlation** of your prediction vs reality. Do your numbers match the reference?

Simulate Your Trades

- Implement a simulation engine:
 - Each day, if your strategy says *buy*, you buy
 - At the end of the day, you *must* sell (no position overnight)
- Track how much cash you have each day
- Simulate the same engine, but using the **true future returns** (a strategy that "knows the future")

Checkpoint

Plot your equity curve over time. What's your ending portfolio value? What's the **Sharpe ratio**?

Add Realism with Fees

- Add
 - A **spread fee** (e.g. 0.1% of trade value)
 - A **fixed fee** per transaction (e.g. \$1).

- Re-run your simulations with fees applied.

Checkpoint

How much lower is your final profit now?

Try Other Strategies

Use the same system to test new strategies. Some ideas:

- **n-day return:** Trade if price change over n days is large enough.
 - $\text{return}(t) = (P(t) - P(t-n)) / P(t-n)$
 - $\text{signal}(t+1) = 1$ if $\text{return}(t) > \theta$, 0 if $\text{return}(t) < -\theta$
- **Moving average crossover:** Buy when a short-term average crosses a long-term one.
 - $\text{MA_fast}(t) = \text{avg}(P(t), \dots, P(t-nf+1))$
 - $\text{MA_slow}(t) = \text{avg}(P(t), \dots, P(t-ns+1))$
 - $\text{signal}(t+1) = 1$ if $\text{MA_fast}(t) > \text{MA_slow}(t)$, else 0
- **Volatility filter:** Trade only when daily price changes are large.
 - $\text{TR}(t) = \max(\text{High-Low}, |\text{High-Close_prev}|, |\text{Low-Close_prev}|)$
 - $\text{ATR}(t) = \text{rolling mean of TR over } n \text{ days}$
 - $\text{signal}(t+1) = 1$ if $P(t) > P(t-1)$ and $\text{ATR}(t)/P(t) > \text{threshold}$
- Re-run your simulations with fees applied.

Checkpoint

Compare strategy results side-by-side. Which is best so far?

Once you've reached this stage, you've completed the minimum project. At the end of the day, we'll give you reference code to compare and improve your version.

Workshop Day 2 – Extend & Optimise Your System

Once you've reached this stage, you've completed the minimum project. At the end of the day, we'll give you reference code to compare and improve your version.

Now you can expand the system and tune it toward the competition. You can choose your path forward depending on your team's interest.

Example Engine Improvements

- Allow **holding positions** (don't sell each day).
- Add **shorting**: bet that the stock will go down (but it costs you to hold).
- Support **multiple stocks** at once.

- Add **risk settings**:
 - Max % per stock
 - Minimum % cash buffer
 - Max % in short positions

Example Strategy Ideas

- More features (e.g. rolling volatility, volume spikes)
- Multiple-step forecasting (e.g. 2 or 3 days ahead)
- Tune parameters like holding length, thresholds, or weights.

Data Hygiene

- Make sure you're not using **future data** in your features or strategy.
- Can you *prove* your strategy is free of look-ahead?

How Will the Best Team Be Chosen?

- Understanding & Explanation
 - Can you clearly explain how your strategy works?
 - Do you know why it behaves the way it does?
- Correctness & Data Hygiene
 - Is your system free from look-ahead errors?
 - Do simulations behave as expected?
- Performance
 - What is your **final portfolio value**?
 - What is your **Sharpe ratio**?
 - Did your strategy generalise across stocks?